GenAI Pathway

Instructor: Samwel Emmanuel

Introduction – Samwel Emmanuel

Data & AI Solutions Architect at Databricks

- Architecting and building multimodal virtual assistants.
- Developing LLM-based agents for Enterprise Data.
- Specializing in cloud-based AI solutions, RAG applications, and AI agent frameworks.

Previously: Data & AI @ Google & Salesforce

Worked on cloud-based AI solutions, data engineering, and enterprise AI applications.

BSc. EECS

Data & AI Solutions Architecture

Teaching Applied ML & GenAI

Welcome to today's class, before we begin, pop into the chat:

Your Name

Designation

Location

Company

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Success Hacks

- Be prepared for the class by watching the warm-up/pre-class videos shared if any
- Attend every live class on Sunday
- Participate in the class
- Practice! Experiment! Become an Expert!!!
- Keep your goal in mind
- Consistency is the key

Be patient with yourself !

IK Support Features

- Uplevel - contains all the learning resources - videos, reading materials, project files
- Discord community for peer learning
- TAs to help answer the queries in the live class
- Use academic support (SSAs) for any support/advise you want
- Support tickets - Raise them any time you want any clarifications/ tech concerns , etc

What happened before?

Evaluation & Agent Orchestration

Evaluating AI Agent

Performance & Accuracy

Orchestration & Multi-Agent Systems

QnA

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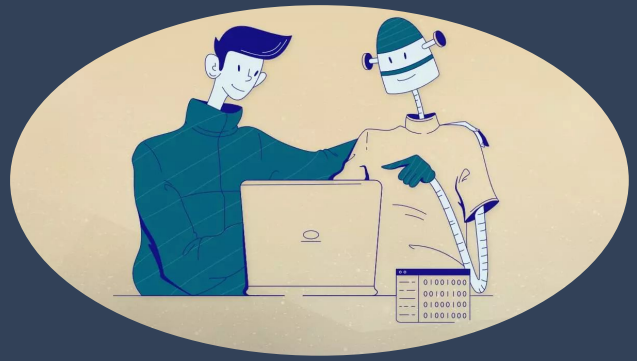
What is going to happen now?

Week 6 of **Self-Led** Breakthrough Project.

You will be driving the project with some assistance from the instructor!!!

Primarily driven through **YOU!**



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Don’t worry!

We are here to help at your every step !

But, how???

IK's Support For You!

We have designed adequate instructional and technical support for you in the form of:

- **Weekend Live Sessions:** Get a picture of your weekly milestones, and get guidance on how to attempt those. Also discuss past progress and get feedback from your instructor and fellow peers. **This is scheduled for every week for the next 8 weeks**

- **1:1 Connects with Project Mentors:** For each project, we have assigned dedicated project mentors to help you with project domain specific queries. To access 1:1 connects, raise a ticket on Uplevel. **You can avail up to 5 1:1 connects in the next 8 weeks.**

- **Technical Coaching Sessions (1:n) :** In this session, you'll be getting your technical doubts resolved from the instructor. They are available **once every 2 weeks (2nd, 4th, 6th and 7th week)**

Structure of sessions for the next 8 weeks!

Week (Today → Week - 4)	Sunday Live Sessions	Coaching Sessions	1<>1 Sessions
Week - 1	✅		You can avail **upto 5 sessions** in these 8 weeks **PRO TIP:** Availing it in the 2nd or 3rd week gives you a good start!!!
Week - 2	✅	✅	
Week - 3	✅		
Week - 4	✅	✅	
Week - 5	✅		
Week - 6	✅	✅	
Week - 7	✅	✅	
Week - 8	✅		

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IK's support extends beyond instructional support...

What more?

- Instructions to plan your **project milestones**
- **Coding notebook templates** for an easy start for every milestone
- **Additional resources** if you get stuck somewhere
- **Milestone completion checklist** (every 2 weeks) to track your progress and know your challenges.
- **AWS credits** for deployment support.
- And much **more...**

Check - Points

01 Week 1st -Week 5th

- ✅ Project Overview
- ✅ Dataset Preparation & Validation
- ✅ Agents Implementations
- ✅ Interim Documentation Review

02 Week 6th -Week 8th

- ✅ Multi-Agent Setup
- ✅ Final Deployment and UI SetUp
- ✅ Final Review and Presentation

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Today's Agenda

Agent Orchestration, Deployment

- Quick Recap and QnA For Week 5
- Orchestration & Multi-Agent Systems
- How to Deploy Your Project
- Project Notebook WalkThrough
- QnA

What happened before?

Evaluation & Agent Orchestration

Evaluating AI Agent

Performance & Accuracy

Orchestration & Multi-Agent Systems

QnA

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How well does your agent generate responses?

Agent Evaluation

- Agent **evaluation** is the process of **assessing** the **performance** and **effectiveness** of Your Agent using **predefined metrics** and **test data**.
- Measures how well the LLM generates **accurate**, **coherent**, and **contextually relevant** responses.
- Tests the model's ability to **handle diverse inputs**, **avoid hallucinations**, and **mitigate biases using human evaluation**, **adversarial testing**, and **factual consistency metrics.**

Orchestration

&

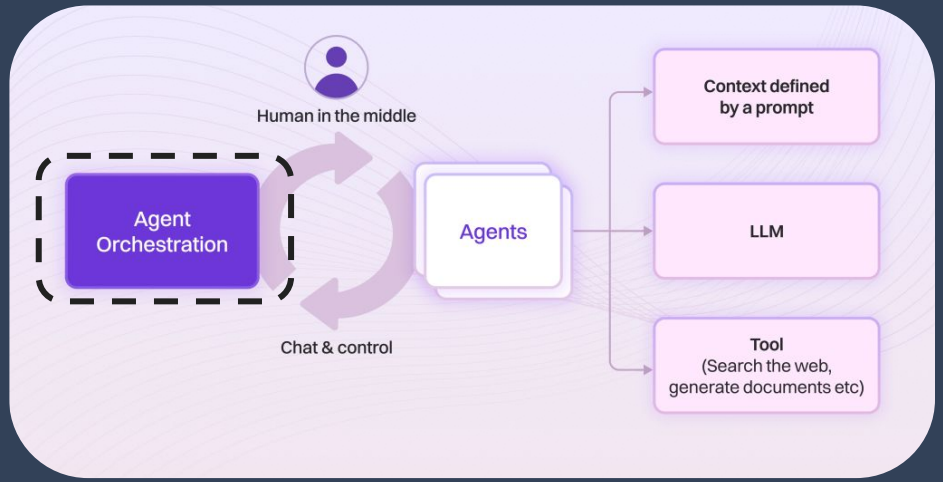
Multi-Agent Systems

What is Agent Orchestration?

Orchestration refers to the **coordination** and management of multiple tasks or agents to work **together** to achieve a complex goal.

- managing **workflows**
- scheduling **tasks**
- ensuring **communication** between different **systems** or **agents**.

Goal: Optimize and **automate** complex **workflows**, enabling efficient **execution of tasks**



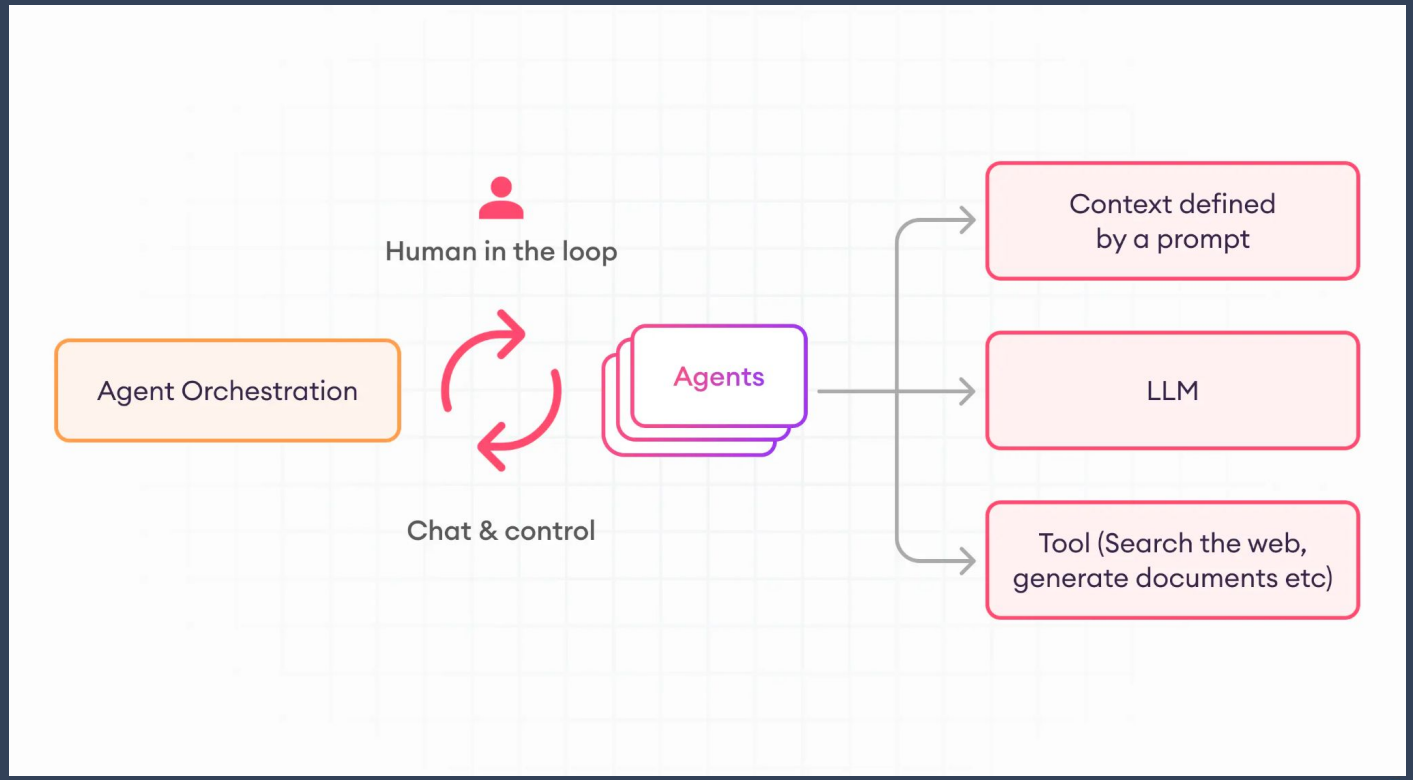
AI Agent Orchestration Steps

AI agent orchestration ensures collaboration, enabling specialized agents to autonomously complete tasks, share data, and optimize workflows.

Key Steps:

- **Assessment & Planning:** Defining goals and identifying requirements.
- **Selection of Specialized Agents:** Choosing the right agents for the tasks.
- **Framework Implementation:** Setting up the orchestration structure.
- **Agent Assignment:** Allocating tasks to appropriate agents.
- **Workflow Coordination:** Managing task execution and dependencies.
- **Data Sharing & Context Management:** Ensuring agents have access to necessary data.
- **Continuous Optimization:** Refining processes for better efficiency.

Multi-Agent System



CrewAI for Multi-Agent Orchestration

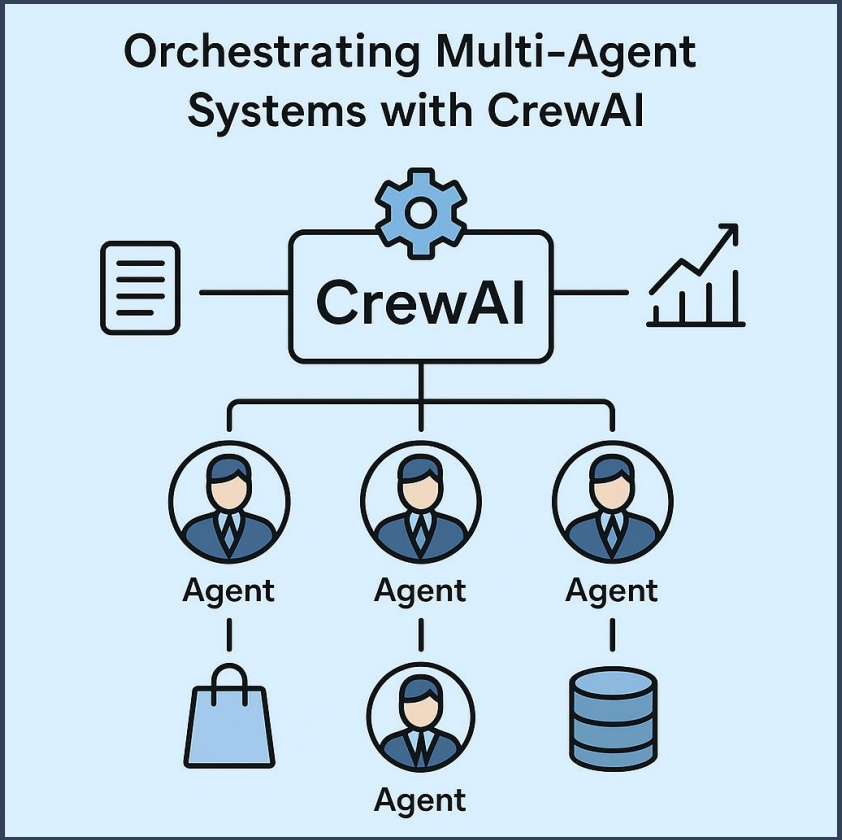
CrewAI is a powerful platform designed for orchestrating multi-agent systems, allowing users to manage and deploy agents with ease.

It offers tools to integrate, automate, and optimize workflows by coordinating multiple agents working in parallel.

Benefits of Using CrewAI for Multi-Agent Orchestration:

- Seamless **Integration**: Easily connects **different agents and systems**.
- **Flexibility**: Customize agent workflows to suit your project needs.
- **Scalability**: Add or remove agents as required for increased complexity or changes in requirements.
- Real-Time Coordination: Ensures agents work in real-time, delivering fast and accurate results.

CrewAI for Multi-Agent Orchestration



Types of Agents

Project	Relevant Agent Used	Type of Agent
CyberShield AI	Threat Intelligence Agent, PII Masking Agent, Multi-Modal Security Agent	**Python Agent, Multi-Modal Agent**
SmartShop AI	Product Recommendation Agent, Price Comparison Agent, FAQ Handling Agent	**Python Agent, SQL Agent**
HealthSense AI	Doctor Availability Agent, Hospital Comparison Agent, Emergency Response Agent	
FinQuery	Financial Query Agent, Automated Insights Agent, Trend Detection Agent	
SalesTrends	Market Trend Analysis Agent, Customer Segmentation Agent, Sales Forecasting Agent	

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Project WalkThroughs

HealthSense AI | Week 6 Milestone

SmartShop AI | Week 6 Milestone

FinQuery/SalesTrend AI | Week 6 Milestone

CyberShield AI | Week 6 Milestone

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Now, Model Evaluation is done!!!

What next????

Making it LIVE.....

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What is Deployment?

Deployment is the process of making your chatbot accessible for real-world use by hosting it on a server or cloud platform. It involves:

- Setting up the backend
- Integrating APIs
- Ensuring that users can interact with the chatbot through:
  - Web app
  - Mobile app
  - Or other interfaces.

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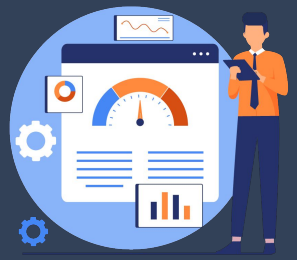
Why do we need it?

Real-World Usage: Allows users to interact with your chatbot beyond your local system.



Performance Testing: Helps you monitor real-time usage, performance, and improvements.

Scalability: Deploying ensures that your chatbot can handle multiple users efficiently.

Portfolio & Experience: Showcasing a deployed project boosts your credibility as a developer.

Deployment Options

Hugging Face

aws

Streamlit

Google Cloud

netlify

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Different Ways to Deploy Your Project

Hugging Face

Streamlit

Covered Today !!!

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Why Hugging Face?

✅ Easy to Deploy: No complex cloud setup—upload your model or app, and it runs.

✅ Free GPU Access: You can run small models for free on Hugging Face Spaces.

✅ Supports Gradio & Streamlit: If your chatbot has a UI, it's plug-and-play.

✅ Community Exposure: Other developers can easily find and use your chatbot.

✅ Good for Prototyping: Test before moving to production-grade cloud deployment.

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Let’s look into the

steps on how to deploy your project

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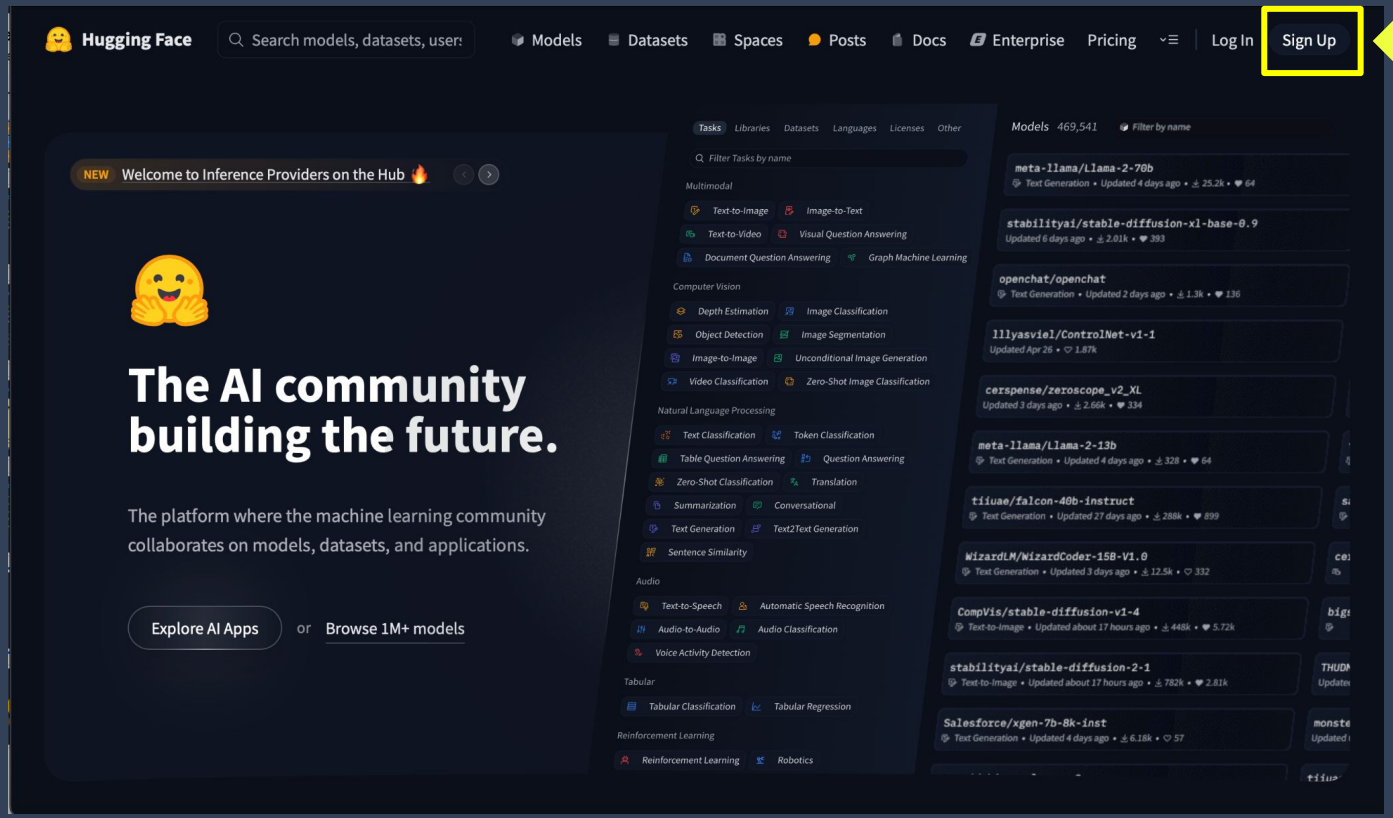
Step 1: Creating your account



Create your hugging face account

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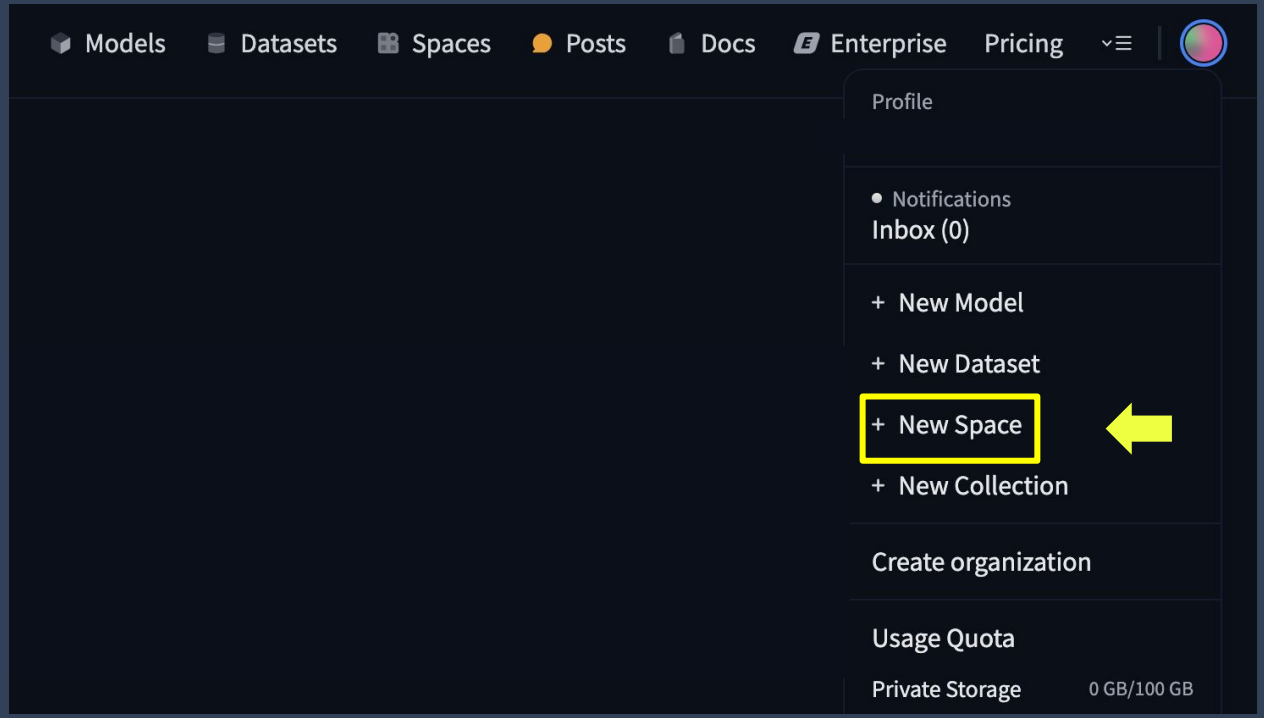
Step 2: Creating a new space



Create a new space

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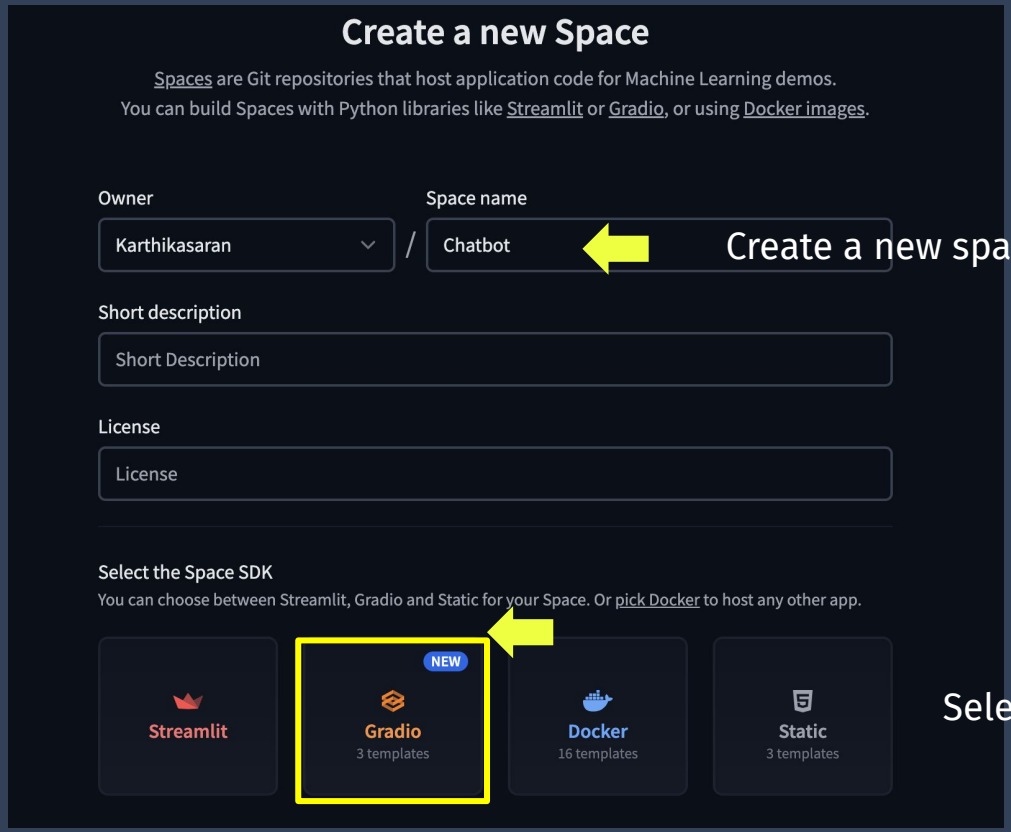

Create a new space

Select Gradio - as we have developed our UI in Gradio

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Step 3: Upload the files

Spaces /Chatbot like 0 No application file

App Files Community Settings

Go to Files

🚀 **Get started with your gradio Space!**

Your new space has been created, follow these steps to get started (or read the full documentation)

Start by cloning this repo by using:

HTTPS SSH

Use an access token as git password/credential

```
# When prompted for a password, use an access token with write permissions.
# Generate one from your settings: https://huggingface.co/settings/tokens
git clone https://huggingface.co/spaces/Karthikasaran/Chatbot
```

Create your Gradio app.py file:

```
import gradio as gr

def greet(name):
    return "Hello " + name + "!!"

demo = gr.Interface(fn=greet, inputs="text", outputs="text")
demo.launch()
```

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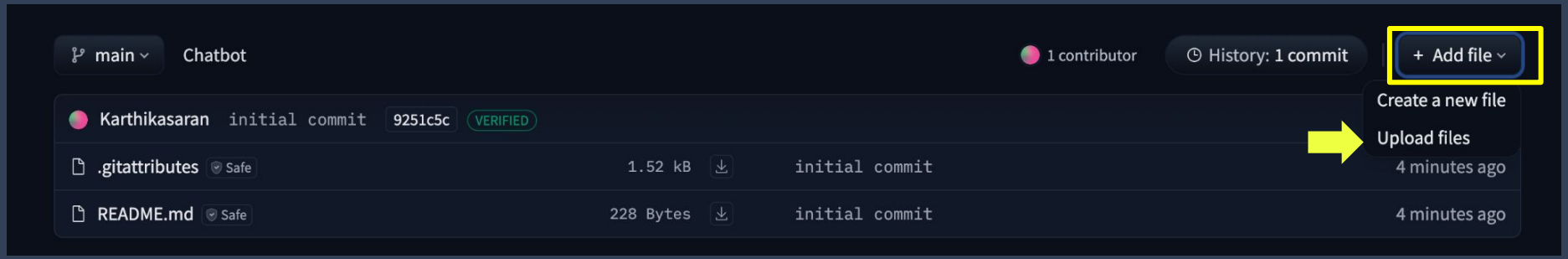
Step 3: Upload the files



Upload the files related to your chatbot

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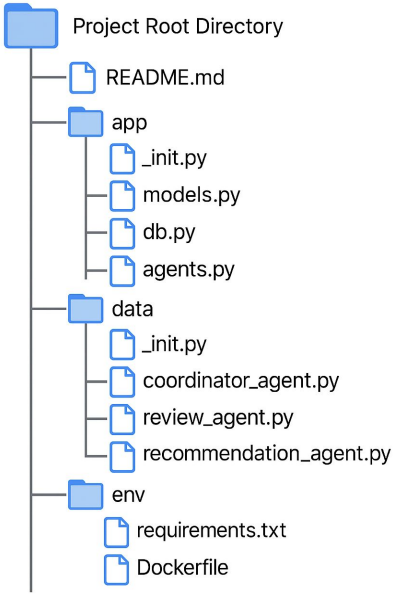
Step 3: Upload the files



Project Structure

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Wait! What are these files?

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Files for Deployment

```
# Installing required packages
!pip install langchainhub
!pip install langchain-openai
!pip install langchain
!pip install beautifulsoup4
!pip install langchain-community
!pip install faiss-cpu
!pip install -U langchain-community tavily-python
!pip gradio_client
!pip install gradio
!pip install langchain_community
```

```
langchain
langchainhub
langchain_community
gradio
gradio_client
tavily-python
faiss-cpu
beautifulsoup4
pandas
numpy
langchain-core
langchain-openai
langchain-text-splitters
langchain-community
faiss-cpu
```

requirements.txt

Add Additional Requirements as Per Your Project

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Shhhh!!!!! Keep your secret, a secret!!!

```
# Setting up Open AI key
os.environ["OPENAI_API_KEY"] = getpass.getpass()

# Setting up the Tavily API Key to do web search
os.environ["TAVILY_API_KEY"] = getpass.getpass()
```

```
## Set the OpenAI API key and model name
Open_API_Key = os.getenv("Open_API_Key")
os.environ["OPENAI_API_KEY"] = Open_API_Key
MODEL="gpt-4o-mini"
client = OpenAI(api_key=Open_API_Key)

## Set the Tavily API key
os.environ["TAVILY_API_KEY"] = os.getenv("Tavily_API_Key")
```

Files for Deployment

.ipynb file

THIS IS YOUR MAIN APPLICATION FILE

app.py file

Now! We will make our Application

more accessible & Scalable

BUT HOW?

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FastAPI + Docker

- FastAPI is used here to handle real-time interactions with our AI models via RESTful APIs.
- This enables external systems (such as a web app or a chatbot UI) to send and receive data to/from the AI model seamlessly.

- Docker wraps the FastAPI app and its dependencies into a container, ensuring portability.
- We'll define the container using a Dockerfile, which includes installation of dependencies, setting environment variables, and launching the application.

Step 4: Writing the FastAPI Code

FastAPI is a modern, fast (high-performance) web framework for building APIs with Python 3.7+ based on standard Python type hints.

It is built on top of Starlette for the web parts and Pydantic for the data parts. FastAPI is designed to be easy to use, flexible, and robust for building APIs.

Why is FastAPI Needed Here?

- **Performance -** FastAPI allows quick API creation with minimal code and handles asynchronous tasks efficiently.
- **Speed -** One of the fastest Python web frameworks, optimized for scaling applications.
- **Automatic Validation -** Automatically validates requests using Pydantic, ensuring proper data formats.
- **Automatic Documentation -** Auto-generates interactive API docs (Swagger UI & ReDoc) for easier testing and consumption.

Step 4: Writing the FastAPI Code

```python
from fastapi import FastAPI
from pydantic import BaseModel
from crewai import Crew, Task

app = FastAPI()

class QueryInput(BaseModel):
    query: str

@app.post("/query")
async def handle_query(input: QueryInput):
    task = Task(description=f"Process: {input.query}",
agent=router_agent)
    crew = Crew(agents=[router_agent, agent_1, agent_2], tasks=[task])
    result = crew.kickoff()
    return {"response": [t.raw for t in result.tasks_output]}
```


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Step 5: Upload the Template UI or Streamlit

Spaces /Chatbot like 0 No application file

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```

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Step 6: Building The DockerFile

Step 1: Start from a Python base image

- Use python:3.11-slim for lightweight and compatibility.

Step 2: Set working directory & copy code

- Copy backend code and your static/ folder containing HTML/CSS.

Step 3: Install Python dependencies

- Include FastAPI, uvicorn, crewai, etc.

Step 4: Expose ports and launch the app

- Use CMD to run the FastAPI app with uvicorn.

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Step 6: Building The DockerFile

```
# Base image
FROM python:3.11.11

# Set working directory
WORKDIR /app

# ✅ Create writable directory for CrewAI storage inside `/src/data`
RUN mkdir -p /src/data && chmod -R 777 /src/data

# Copy all project files
COPY . .

# Install dependencies
RUN pip install --no-cache-dir -r requirements.txt

# Expose FastAPI Port
EXPOSE 7860

# Run FastAPI
CMD ["uvicorn", "src.app:app", "--host", "0.0.0.0", "--port",
"7860"]
```

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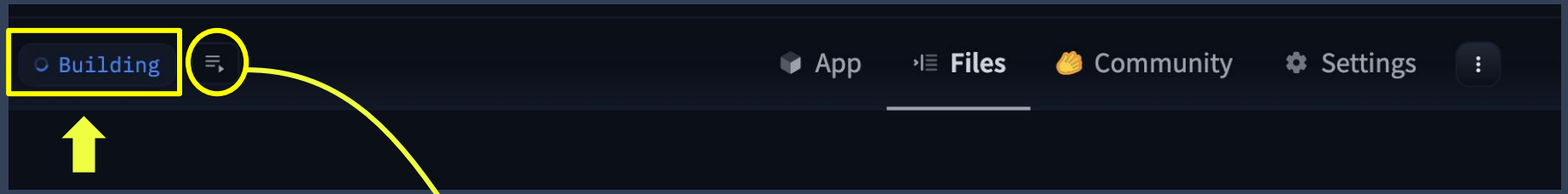
Done Uploading!!!!

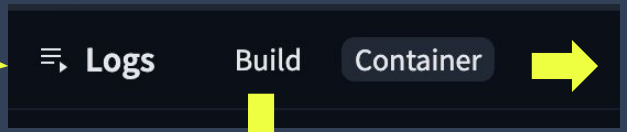
What next?

Step 7: Building the Application



After uploading app.py file, automatic building of your application happens.

Check the logs here



Logs while running the app

Logs while building the app

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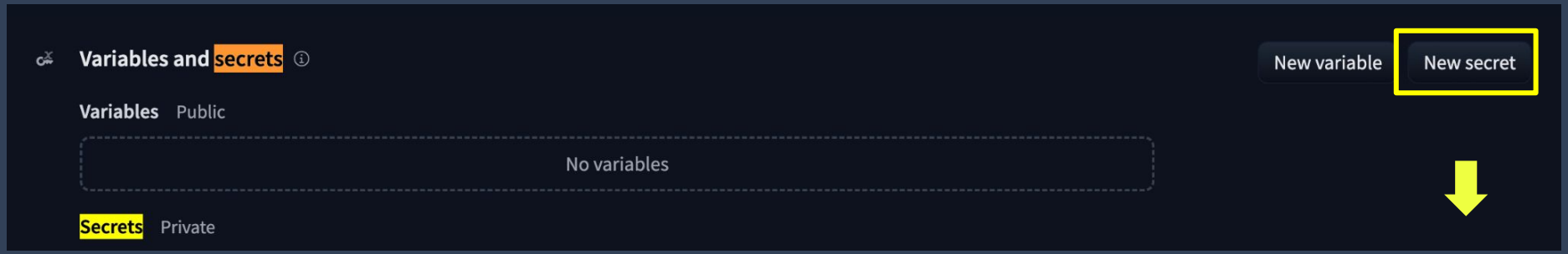
Did you secretly share your

API Keys????

Sharing the API keys secretly

Search for ‘secrets’



Create the ‘New Secret’ for

Open AI key & Tavily key

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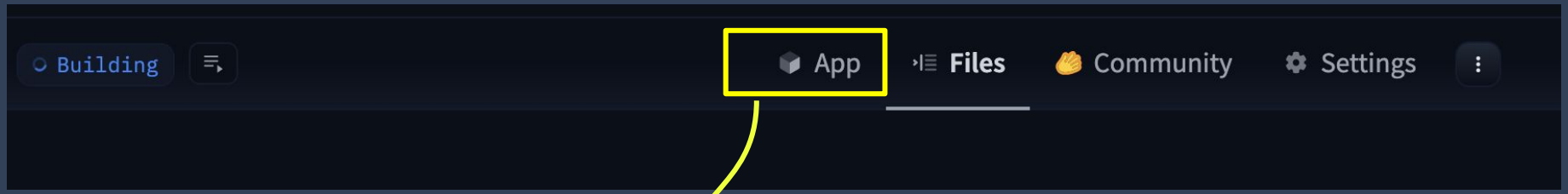
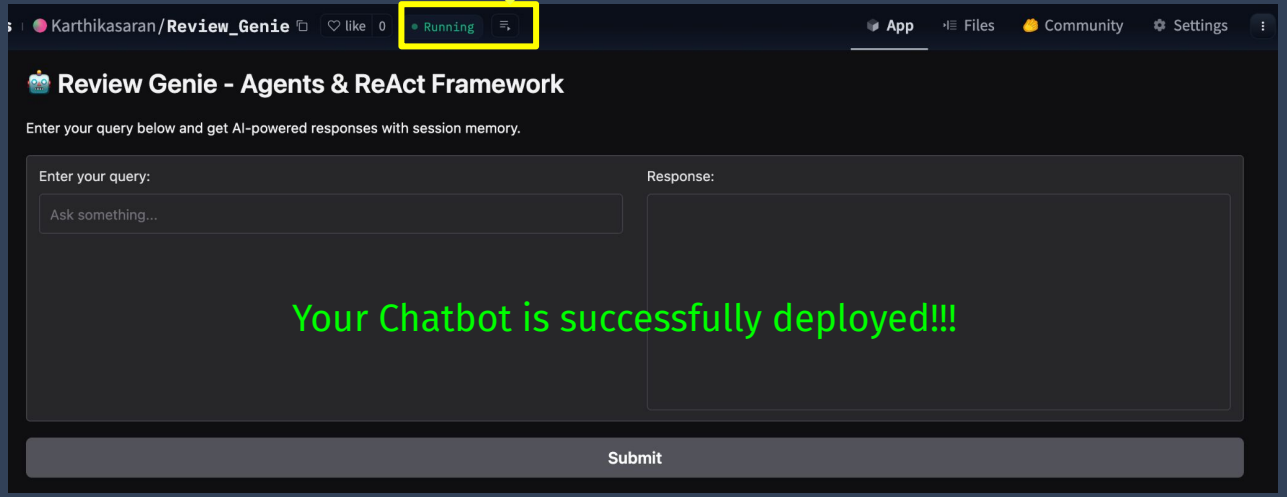
Step 8: Running the Application



Application Status



Your Chatbot is successfully deployed!!!

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Time for Deployment!!!

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Project WalkThroughs

HealthSense AI | Week 6 Milestone

SmartShop AI | Week 6 Milestone

FinQuery/SalesTrend AI | Week 6 Milestone

CyberShield AI | Week 6 Milestone

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Now! Let’s do the same with

Streamlit Cloud

BUT HOW?

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Streamlit Cloud

Streamlit is a fast and easy-to-use tool for building interactive web applications and dashboards in Python, particularly designed for machine learning and data science projects.

Streamlit

What does it do?

- Streamlit allows you to quickly turn Python scripts into beautiful, real-time web applications.
- It's ideal for showcasing machine learning models, data visualizations, or any interactive tools. It works seamlessly with libraries like FastAPI and Docker to create interactive user interfaces for AI models.

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Streamlit Cloud

Why use Streamlit?

- Quick UI creation: It allows you to build web-based applications in minutes.
- Real-time updates: It enables instant interaction with users, making it perfect for deploying AI/ML models and showcasing them in a dynamic UI.
- Easy integration: Streamlit integrates smoothly with FastAPI to connect with backend services and make models accessible.

Streamlit

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Now! Let’s go through

Step-by-Step to

Deploy on Streamlit Cloud

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Project WalkThroughs

HealthSense AI | Week 6 Milestone

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